

Widad Dabbas

☎ +44756314511 | ✉ dabbaswidad@gmail.com | 🔗 linkedin.com/in/widad-dabbas/

Education

Durham University

MSc in Mathematical Sciences

Durham, UK

September 2024 - Present

- **Proposed Thesis:** Developing ecological complexity-stability theory for the soil microbiome under the presence of heavy metals through mathematical models and novel data methodology

University of St Andrews

BSc in Biology and Mathematics with Honours

St Andrews, UK

Sept 2019 - June 2023

- Graduated with merit
- Kendo Club Safety Officer
- **Biology Dissertation:** Disturbance and Niche Construction as Mechanisms of Community Assembly
- **Mathematics Dissertation:** Category Theory for Distinguishing Knot Invariants

Experience

Soul Curated Wellness

Regenerative Design Consultant

Amman, Jordan

January - August 2024

- Consulted on design plans, delivering research and solutions of regenerative materials and processes.
- Examined local ecosystems through on site and data analyses to inform mutualistic design opportunities.
- Lead a team of 4 to produce a comprehensive master-plan including presentation, realistic timeline and budget.

Control Theory and Systems Biology Lab, ETH Zurich

Visiting Student

Basel, Switzerland

August - September 2022

- Used MATLAB to simulate controllers in discrete chemical reaction networks.
- Assisted in molecular cloning to build the theorised biological controllers in yeast cells.
- Investigated the dynamical properties of simulated networks and constructed controllers.

Royal Botanic Garden

Assistant Botanist

Balqa, Jordan

January - June 2022

- Utilized irisBG GIS technology to map and analyze plant populations.
- Preserved seed bank and herbarium specimens whilst maintaining up to date databases.
- Gained expertise in seed and vegetative pretreatments, propagation and plant care methods and sediment preparation.

Girguis Lab, Harvard University

Laboratory Intern

Cambridge, MA, USA

September - December 2018

- Independently constructed an OpenROV and used it to gather underwater footage in the Gulf of Aqaba.
- Collaborated on a video research proposal as a director, editor and translator, making use of ROV footage.
- Shadowed PhD research on *Sporomusa ovata*, and was responsible for independent work with anaerobic liquid cultures.

Ocean Exploration Trust, E/V Nautilus

Science/Data Team and Video Engineer

Pacific Ocean, CA, USA

July - August 2017

- Remotely operated the 23 live broadcast cameras on ROVs Hercules and ARGUS and aboard the E/V Nautilus.
- Engaged in live radio discussion and analysis of underwater footage with the science team.
- Prepared deep sea samples in the onboard wet lab for further use by onshore researchers.

Projects

Earth's Whispers: Bridging Soil Science and Fine Arts

Jordan National Gallery of Fine Arts

Amman, Jordan

August 2024

Curated my debut solo exhibition as a first example of integrating science with fine arts in Jordan. Inspired by soil as a vital ecosystem and an impressive natural material, I played with the dichotomy of glass prints on mud sculptures. Brief microbial and structural analysis was also conducted in collaboration with professors at the University of Jordan to produce a complimentary film.

Financial Time Series Topological Data Analysis (TDA)

Aalborg University

Aalborg, Denmark

June 2023

Investigated and presented on the predictive capabilities of TDA as part of the Danish-Swedish summer school on Topological Data Analysis and spatial statistics. Conducted TDA analyses with reference to financial time series and historical records using Python. No conclusive results were made, but persistent entropy appeared to be negatively correlated with market volatility.

Experimental Research Project: Ecological Niche Construction

St Andrews, UK

University of St Andrews

September 2022 - April 2023

Conducted research into the effects of ecological niche construction under various disturbance regimes under climate change and severe weather events. Research was primarily conducted by computational methods adapted from the literature. Links were then drawn from the defined ecological model to the existing knowledge and data from with coral reef ecosystems in mind.

Project in Pure Mathematics: Knot invariants and Categorification

St Andrews, UK

University of St Andrews

September 2022 - April 2023

Built up to the Khovanov homology as a justification of applied category theory. Independent study of category theory, homological algebra and knot theory was undertaken. Care was taken to introduce complex topics to the reader through written or diagrammatic means. A strong narrative was achieved and carried through technical definitions and reimagined proofs.

Intertidal Project on *Ascophylum nodosum*

St Andrews, UK

University of St Andrews

August - September 2023

Designed and conducted an observational study relating *A. nodosum* length to shoreline elevation. Line-intercept sampling was implemented to collect data. ANOVA and Akaike information criterion (AIC) were applied in R for data and statistical analyses. Results negatively correlated *A. nodosum* length with shoreline elevation irrespective of age.